Congruence Classes, Units, and the Arithmetic of Exponents

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§1.1 Congruence is equality in a quotient ring

For $n \geq 2$, the relation

$$a \equiv b \pmod{n} \quad \Longleftrightarrow \quad n \mid (a-b)$$

partitions $\mathbb{Z}$ into n classes. It is an equivalence relation and respects the two ring operations:

$$a \equiv b, \quad c \equiv d \pmod{n} \implies a+c \equiv b+d, \quad ac \equiv bd \pmod{n}.$$

The residue class $[a]$ is the coset $a + n\mathbb{Z}$, and the quotient

$$\mathbb{Z}/n\mathbb{Z}$$

is a ring. A complete residue system consists of one representative from each of its classes. For example, $0, 1, \ldots, n-1$ is one such system, but replacing any representative by a congruent integer leaves it complete.

The purpose of reducing a complicated expression is therefore algebraic: evaluating a polynomial of integers and then passing to the quotient gives the same answer as reducing every integer first. Modular arithmetic is the image of integer arithmetic under a ring homomorphism, which explains why these reductions compose reliably. For the number-theoretic organization we follow Abhinav Kumar's MIT 18.781 course notes on congruences, Euler's theorem, the Chinese remainder theorem, and primitive roots.[1]

Example 1.1.1 (Three ways to avoid multiplying large numbers)
The last digit of 15^{200} is 5, because $15^r \equiv 5 \pmod{10}$ for every positive integer r. Modulo 17,

$$15^{100} \equiv (-2)^{100} = 2^{100}, \qquad 2^4 \equiv -1, \quad 2^8 \equiv 1,$$

so $2^{100} \equiv (2^8)^{12} 2^4 \equiv 16 \pmod{17}$. Finally,

$$\begin{aligned} &2001 \cdot 2002 + 2003 \cdot 2004 + 2005 \cdot 2006 \\ &\quad \equiv 21 \cdot 22 + 23 \cdot 24 + 25 \cdot 26 \\ &\quad \equiv 12 + 12 + 20 \equiv 44 \pmod{45}. \end{aligned}$$

The examples illustrate reduction of a constant base, reduction by a multiplicative period, and reduction before distributing products.

Quadratic residues provide another useful finite check. Since every integer is congruent to $0, \ldots, 7 \pmod{8}$, squaring gives

$$\{x^2 \bmod 8 : x \in \mathbb{Z}\} = \{0, 1, 4\}.$$

[1] https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/

The eight possibilities can be reduced further: an odd integer has square $1 \pmod 8$; an even integer is either divisible by 4, giving square $0 \pmod 8$, or is $2 \pmod 4$, giving square $4 \pmod 8$.

§1.2 Reduced residue systems form the group of units

A class $[a] \in \mathbb{Z}/n\mathbb{Z}$ is invertible if and only if $\gcd(a, n) = 1$. Indeed, Bézout's identity $ua + vn = 1$ supplies an inverse of $[a]$, whereas an inverse modulo n implies $\gcd(a, n) = 1$.

Consequently the reduced residue classes form the multiplicative group

$$(\mathbb{Z}/n\mathbb{Z})^\times.$$

Its cardinality is Euler's totient

$$\varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\times.$$

For $n = p^r$, precisely the multiples of p are excluded, so

$$\varphi(p^r) = p^r - p^{r-1}.$$

For coprime m, n, the Chinese remainder theorem gives a ring isomorphism

$$\mathbb{Z}/mn\mathbb{Z} \simeq \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z},$$

which restricts to the unit groups. Their sizes multiply:

$$\varphi(mn) = \varphi(m)\varphi(n).$$

Factoring $n = \prod_{p|n} p^{v_p(n)}$ now derives

$$\boxed{\varphi(n) = n \prod_{p|n} (1 - p^{-1}).}$$

This is a deduction from counting units and the CRT, not an unrelated formula to memorize.

§1.3 Exponent reduction is a statement about element order

For a unit $a \pmod n$, define its multiplicative order by

$$\operatorname{ord}_n(a) = \min\{d \geq 1 : a^d \equiv 1 \pmod n\}.$$

The finite-group theorem of Lagrange implies

$$\operatorname{ord}_n(a) \mid \varphi(n), \qquad a^{\varphi(n)} \equiv 1 \pmod n.$$

This is Euler's theorem; it requires $\gcd(a, n) = 1$. A much smaller order may exist. For example,

$$2^4 \equiv -1 \pmod{17}, \quad 2^8 \equiv 1 \pmod{17},$$

so $\operatorname{ord}_{17}(2) = 8$, whereas $\varphi(17) = 16$. The exponent 100 can be reduced modulo 8, the actual group order of the cyclic subgroup generated by 2. By contrast, $15 \pmod{10}$ is not a unit, so applying Euler's theorem to the last-digit problem would be invalid.

Consider the sequence

$$F_k = 12^{2k+1} + 36^k + 5^k + 3, \qquad k \geq 1.$$

Every term has $F_k \equiv 0 \pmod 4$. A finer residue computation gives

$$F_1 \equiv 12 \pmod{16}, \qquad F_k \equiv 5^k + 3 \pmod{16} \quad (k \geq 2),$$

because $12 \cdot 144^k$ and 36^k are divisible by 16 for $k \geq 2$. The powers $5^k \pmod{16}$ cycle with period four:

$$5, \quad 9, \quad 13, \quad 1.$$

It follows that

$$\boxed{16 \mid F_k \quad \Longleftrightarrow \quad k \equiv 3 \pmod 4} \qquad (k \geq 1).$$

Reducing to a small multiplicative cycle exposes more structure than checking isolated numerical values; it also shows why the precise modulus is central to a divisibility question.

§1.4 Primitive roots give discrete-logarithm coordinates

A primitive root modulo n is a generator of the unit group. If $(\mathbb{Z}/n\mathbb{Z})^\times = \langle g \rangle$, then every unit has the form g^j; multiplication becomes addition of exponents modulo $\varphi(n)$. For odd primes p, the unit group modulo p^r is cyclic, and a generator exists. For $n = 8$, however,

$$(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$$

and all nonidentity elements have square 1, so the group is not cyclic.

Apart from the trivial modulus 1, the classical classification is that primitive roots exist precisely for

$$n = 2, \ 4, \ p^r, \ 2p^r$$

with p an odd prime. A convenient primitive root is thus not available for every modulus. In other cases the Chinese remainder theorem and the structure theorem for finite abelian groups are the better language.

§1.5 Fourier Theory of the Unit Group

A Dirichlet character modulo n is a homomorphism

$$\chi : (\mathbb{Z}/n\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times,$$

extended to all integers by 0 on nonunits. Since the unit group is finite, the nonzero values of χ are roots of unity. If G is a finite abelian group, its characters satisfy

$$\sum_{\chi \in \widehat{G}} \chi(g) = \begin{cases} |G|, & g = e, \\ 0, & g \neq e. \end{cases}$$

To see the cancellation in the second line, choose a character ψ with $\psi(g) \neq 1$. Multiplying each summand by $\psi(g)$ merely permutes the characters, so the sum equals $\psi(g)$ times itself and must vanish.

For $G = (\mathbb{Z}/n\mathbb{Z})^\times$, this yields an indicator for a specified invertible congruence class:

$$\mathbf{1}_{a\equiv b \pmod n} = \frac{1}{\varphi(n)} \sum_{\chi \pmod n} \chi(a)\overline{\chi(b)}$$

when both a, b are units modulo n. This is the exact connection between the original residue-class exercises and Fourier analysis on a finite group. It is also why characters enter the study of primes in arithmetic progressions.